

Deep Q-Network (DQN): Architecture & Core Concepts

DQN represents the fusion of **Q-Learning** with **Deep Neural Networks**, effectively overcoming the instability of function approximation in non-linear settings through two main mechanisms: **Experience Replay** and **Target Networks**.

1. Core Architecture

The DQN agent uses a neural network (the **Policy Network**) to approximate the action-value function $Q(s, a; \theta)$.

- **Input:** State representation s (e.g., raw pixels or feature vectors).
- **Output:** A vector of Q-values $Q(s, a_i)$ for every possible action $a_i \in \mathcal{A}$.
 - *Advantage:* A single forward pass computes values for all actions, enabling efficient greedy action selection: $a^* = \arg \max_a Q(s, a)$